Experiment 5 - Installing Jenkins

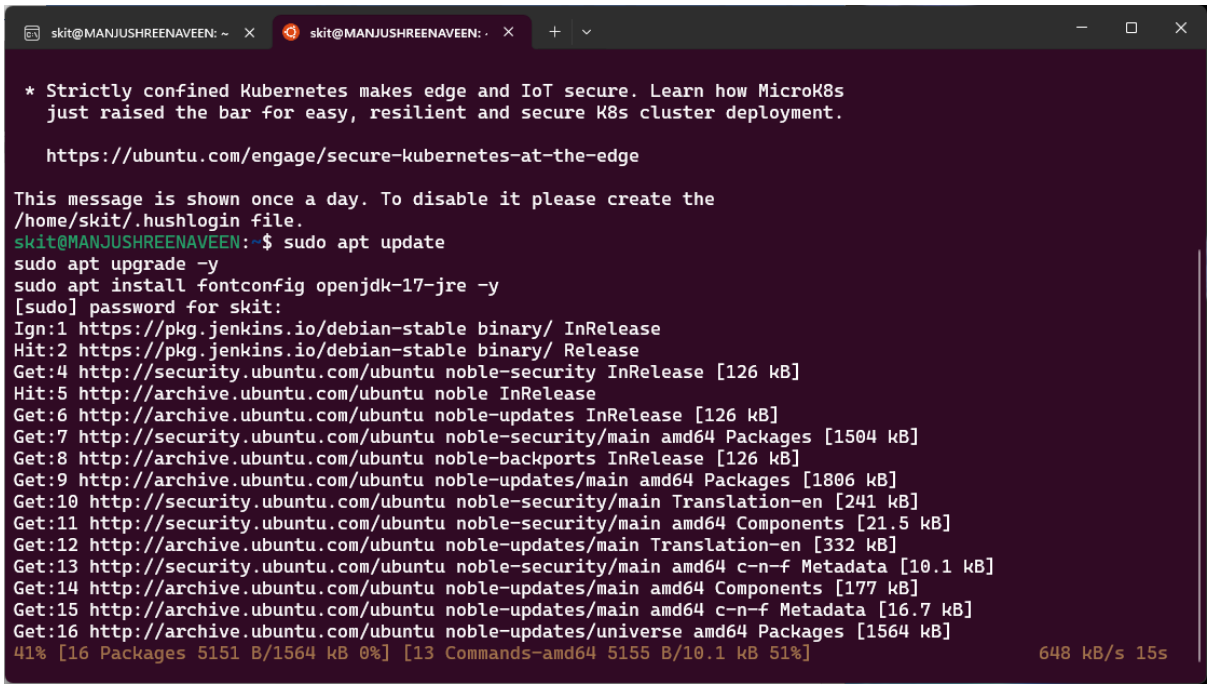
Important: Jenkins requires a compatible version of Java. Ensure you have **Java 17** installed on your system before proceeding with the steps below.

1. Update Your System

Open your terminal and ensure your package list is updated and upgrades are applied:

```
sudo apt update
sudo apt upgrade -y
```

sudo apt install fontconfig openjdk-17-jre -y



```
skit@MANJUSHREENAVEEN: ~    skit@MANJUSHREENAVEEN: ~

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/skit/.hushlogin file.
skit@MANJUSHREENAVEEN:~$ sudo apt update
sudo apt upgrade -y
sudo apt install fontconfig openjdk-17-jre -y
[sudo] password for skit:
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1504 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1806 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [241 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [332 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.1 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1564 kB]
41% [16 Packages 5151 B/1564 kB 0%] [13 Commands-amd64 5155 B/10.1 kB 51%]          648 kB/s 15s
```

java -version

```
skit@MANJUSHREENAVEEN: ~   skit@MANJUSHREENAVEEN: ~
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  libnss3
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
1 standard LTS security update
Need to get 1445 kB of archives.
After this operation, 4096 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss3 amd64 2:3.98-1ubuntu0.1 [1445 kB]
Fetched 1445 kB in 2s (603 kB/s)
(Reading database ... 55773 files and directories currently installed.)
Preparing to unpack .../libnss3_2%3a3.98-1ubuntu0.1_amd64.deb ...
Unpacking libnss3:amd64 (2:3.98-1ubuntu0.1) over (2:3.98-1build1) ...
Setting up libnss3:amd64 (2:3.98-1ubuntu0.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fontconfig is already the newest version (2.15.0-1.1ubuntu2).
openjdk-17-jre is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
skit@MANJUSHREENAVEEN:~$ java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
skit@MANJUSHREENAVEEN:~$
```

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2. Add the Jenkins Repository Key

Download and add the GPG key for the Jenkins repository:

sudo wget -O /usr/share/keyrings/jenkins-keyring.asc https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key

```
skit@MANJUSHREENAVEEN: ~   skit@MANJUSHREENAVEEN: ~
Preparing to unpack .../libnss3_2%3a3.98-1ubuntu0.1_amd64.deb ...
Unpacking libnss3:amd64 (2:3.98-1ubuntu0.1) over (2:3.98-1build1) ...
Setting up libnss3:amd64 (2:3.98-1ubuntu0.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fontconfig is already the newest version (2.15.0-1.1ubuntu2).
openjdk-17-jre is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
skit@MANJUSHREENAVEEN:~$ java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
skit@MANJUSHREENAVEEN:~$ sudo wget -O /usr/share/keyrings/jenkins-keyring.asc https://pkg.jenkins.io/debian-
stable/jenkins.io-2026.key
--2026-03-09 06:41:12--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.2.133, 151.101.130.133, 151.101.66.133, ...
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.2.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/usr/share/keyrings/jenkins-keyring.asc'

/usr/share/keyrings/jenkin 100%[=========================================>]   1.64K  --.-KB/s    in 0s

2026-03-09 06:41:12 (8.29 MB/s) - '/usr/share/keyrings/jenkins-keyring.asc' saved [1680/1680]

skit@MANJUSHREENAVEEN:~$
```

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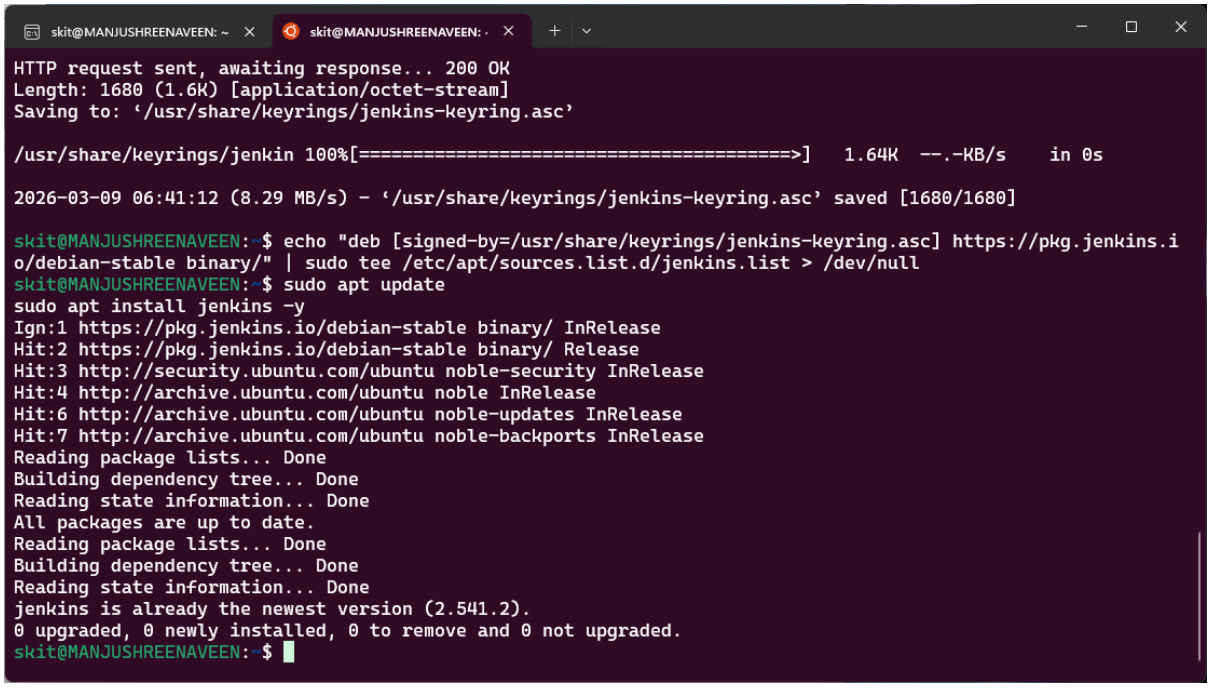
3. Add the Jenkins Repository to Your Sources List

Add the Jenkins stable repository to your system's software sources:

echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null



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4. Update Your Package List and Install Jenkins

Update your package list again to include the Jenkins repository, then install Jenkins:

```
sudo apt update
sudo apt install jenkins -y
```

```
skit@MANJUSHREENAVEEN: ~    skit@MANJUSHREENAVEEN: ·
/usr/share/keyrings/jenkin 100%[=====================================>]   1.64K  --.-KB/s    in 0s

2026-03-09 06:41:12 (8.29 MB/s) - '/usr/share/keyrings/jenkins-keyring.asc' saved [1680/1680]

skit@MANJUSHREENAVEEN:~$ echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.i
o/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
skit@MANJUSHREENAVEEN:~$ sudo apt update
sudo apt install jenkins -y
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jenkins is already the newest version (2.541.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
skit@MANJUSHREENAVEEN:~$ sudo systemctl start jenkins
sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
skit@MANJUSHREENAVEEN:~$
```

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5. Start and Enable the Jenkins Service

Start the Jenkins service and configure it to launch automatically on system boot:

```
sudo systemctl start jenkins
sudo systemctl enable jenkins
```

Check the status to confirm it's running:

```
sudo systemctl status jenkins
```

Your output should be similar to this:

```
skit@MANJUSHREENAVEEN: ~   skit@MANJUSHREENAVEEN: ~
jenkins is already the newest version (2.541.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
skit@MANJUSHREENAVEEN:~$ sudo systemctl start jenkins
sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
skit@MANJUSHREENAVEEN:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-09 06:39:49 UTC; 3min 8s ago
   Main PID: 213 (java)
      Tasks: 45 (limit: 3443)
     Memory: 649.1M (peak: 650.6M)
        CPU: 1min 11.368s
     CGroup: /system.slice/jenkins.service
             └─213 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/c>

Mar 09 06:39:48 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:39:48.753+0000 [id=42]        INFO        jenk>
Mar 09 06:39:48 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:39:48.795+0000 [id=39]        INFO        jenk>
Mar 09 06:39:48 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:39:48.847+0000 [id=63]        INFO        huds>
Mar 09 06:39:49 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:39:49.004+0000 [id=45]        INFO        jenk>
Mar 09 06:39:49 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:39:49.174+0000 [id=33]        INFO        huds>
Mar 09 06:39:49 MANJUSHREENAVEEN systemd[1]: Started jenkins.service - Jenkins Continuous Integration Serve>
Mar 09 06:40:02 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:40:02.181+0000 [id=63]        INFO        h.m.>
Mar 09 06:40:04 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:40:04.326+0000 [id=63]        INFO        h.m.>
Mar 09 06:40:06 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:40:06.447+0000 [id=63]        INFO        h.m.>
Mar 09 06:40:06 MANJUSHREENAVEEN jenkins[213]: 2026-03-09 06:40:06.448+0000 [id=63]        INFO        huds>
lines 1-20/20 (END)
```

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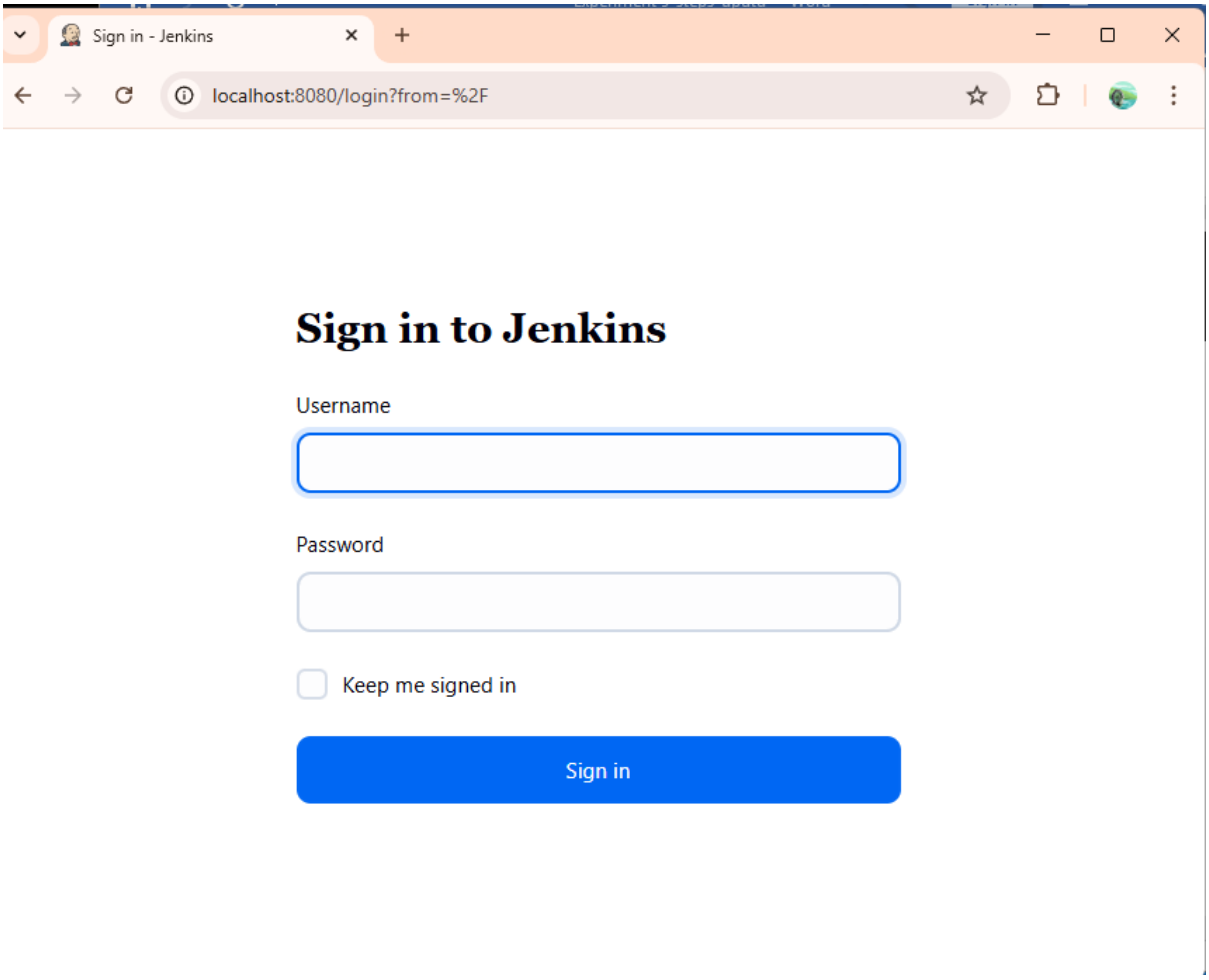
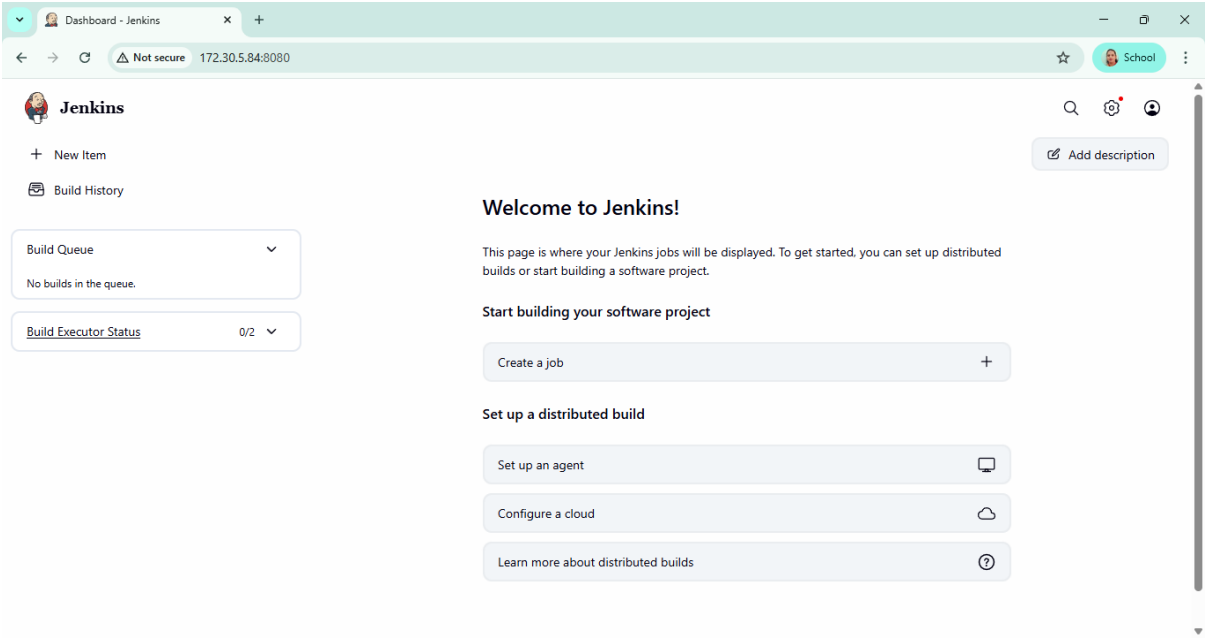
6. Access Jenkins and Complete Initial Setup

1. Open a web browser and navigate to the Jenkins default port:

   localhost:8080

2. You will be prompted to unlock Jenkins. Retrieve the initial administrator password by running the following command in your terminal:
3. `sudo cat /var/lib/jenkins/secrets/initialAdminPassword`
4. Copy the displayed password and paste it into the "Administrator password" field in your browser. Click "Continue".
5. On the next screen, click "Install suggested plugins" to install a recommended set of plugins.
6. After the plugins are installed, you will be asked to create an administrator user. Fill in the required details and click "Save and Finish".
7. You should now see the Jenkins dashboard, ready for use.


Sign in - Jenkins
localhost:8080/login?from=%2F
Sign in to Jenkins
Username
Password
Keep me signed in
Sign in


Dashboard - Jenkins
Not secure
172.30.5.84:8080
School
Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2
Add description
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
Learn more about distributed builds